

Mangrove Extent Change and Blue Carbon Loss in Southeast Asia

A regional analysis of mangrove dynamics and carbon emissions across nine countries using Global Mangrove Watch v3.0 (1996–2020)

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Tools: GEE · R · QGIS · Python

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1. Introduction

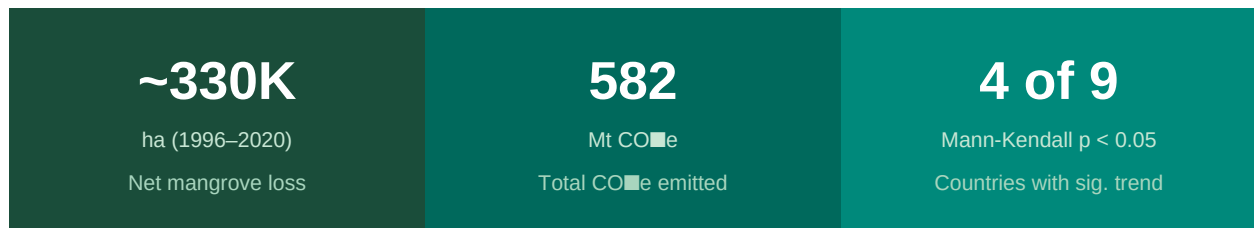
Mangrove forests occupy the intertidal zone across tropical and subtropical coastlines, and Southeast Asia holds the largest and most species-rich concentration on the planet. The region accounts for roughly one-third of global mangrove extent and supports more than 50 species of mangrove-forming plants (Spalding et al., 2010; Giri et al., 2011). Beyond biodiversity, these ecosystems deliver services on which millions of coastal communities depend, from fisheries nursery habitat and shoreline protection to fuelwood and food security (Hochard et al., 2019; Primavera et al., 2019).

One property that has placed mangroves at the centre of climate policy discussions is their exceptional capacity to store carbon. Mangrove soils accumulate organic matter over centuries under anoxic conditions, making whole-ecosystem carbon densities several times higher than those of upland tropical forests (Donato et al., 2011). When mangroves are cleared, this stored carbon is mineralised and released as CO₂, contributing to greenhouse gas emissions that are disproportionately large relative to the ecosystem's areal extent. Pendleton et al. (2012) estimated that global mangrove deforestation releases between 0.02 and 0.12 Pg of CO₂ equivalent per year, equivalent to 2 to 8 percent of emissions from tropical deforestation despite mangroves covering less than one percent of the global forest area. These losses are framed under the concept of blue carbon, the organic carbon stored and sequestered by vegetated coastal ecosystems including mangroves, seagrasses, and salt marshes.

Deforestation pressure in Southeast Asia has been severe and sustained. Richards and Friess (2016) documented the loss of more than 100,000 ha of mangrove across the region between 2000 and 2012, identifying aquaculture expansion, rice agriculture in Myanmar, and oil palm plantations in Malaysia and Indonesia as the principal proximate drivers. Rahman et al. (2024) extended that analysis through 2020 using Global Mangrove Watch v3.0 (GMW v3.0; Bunting et al., 2022), reporting a decline from approximately 5.07 Mha in 1996 to 4.82 Mha by 2020 and estimating associated CO₂ emissions for all countries in the region.

The present report builds on those contributions. It covers the same period and data source as Rahman et al. (2024) but applies two methodological additions that are not present in prior regional assessments. First, it uses the Mann-Kendall non-parametric trend test and Sen's Slope estimator to formally quantify whether national-level mangrove area time series show statistically significant directional trends over 1996 to 2020, an analysis that prior studies describe narratively but do not test statistically. Second, it calculates blue carbon emissions using country-specific emission factors from Hamilton and Friess (2018), which are calibrated against IPCC Wetlands Supplement standards and reflect the fraction of the carbon stock that is actually mineralised upon deforestation, rather than treating the entire standing stock as immediately emitted. The analysis covers nine countries: Indonesia, Myanmar, Malaysia, Thailand, the Philippines, Cambodia, Vietnam, Brunei Darussalam, and Timor-Leste. Singapore is excluded because its mangrove extent falls below 1,000 ha and its dynamics are driven primarily by urban land reclamation rather than the land-use pressures characteristic of the broader region.

This report is intended as a technical complement to the body of research produced by Richards and Friess (2016), Hamilton and Friess (2018), and Rahman et al. (2024). It does not seek to challenge or replace those findings, which carry greater analytical depth and validation resources. The goal is to extend the temporal coverage, add a formal statistical layer, and present the results in a format accessible to conservation practitioners and environmental managers working in the region.

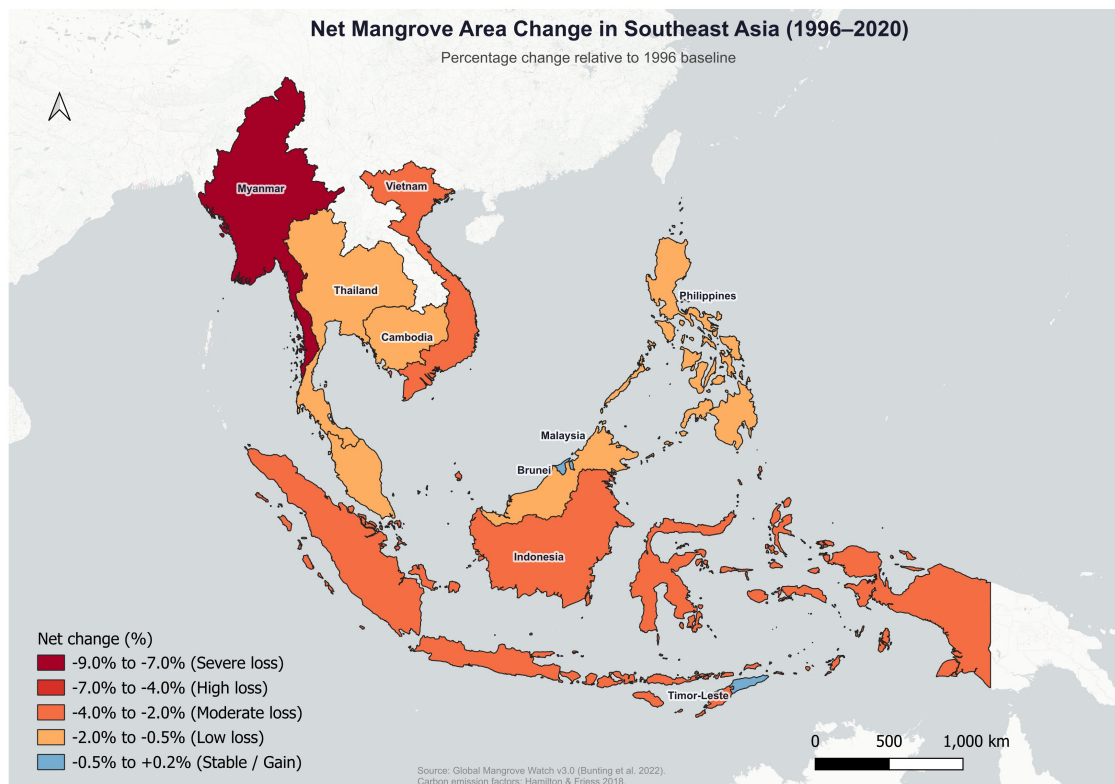


2. Study Area

The study covers nine countries in mainland and insular Southeast Asia: Indonesia, Myanmar, Malaysia, Thailand, the Philippines, Cambodia, Vietnam, Brunei Darussalam, and Timor-Leste. Together these countries contained approximately 5.0 million hectares of mangrove in 1996, representing the largest continuous concentration of mangrove forest on Earth (Spalding et al., 2010). The region spans the Indo-Pacific biogeographic zone, characterised by maximum mangrove species richness and among the highest carbon densities recorded globally (Donato et al., 2011; Sharma et al., 2023).

Mangrove distribution across the region is highly uneven. Indonesia alone held approximately 3.07 Mha in 1996, representing 61 percent of the regional total, concentrated in Kalimantan, Sumatra, Papua, and Sulawesi. Myanmar contributes the second largest extent (0.58 Mha), dominated by the Ayeyarwady Delta and the Rakhine and Tanintharyi coasts. Malaysia (0.54 Mha) holds extensive mangroves on the Malay Peninsula and in Sabah and Sarawak on Borneo. The remaining six countries each contribute less than 0.30 Mha, with Brunei and Timor-Leste representing the smallest extents in the dataset.

The region is characterised by two broad coastal geomorphologies that influence mangrove carbon storage. Estuarine and deltaic systems, such as the Ayeyarwady Delta in Myanmar and the Mekong Delta in Vietnam, accumulate organic matter under low-energy, reducing conditions and tend to support high soil carbon stocks. Open-coast and island-fringe mangroves, more common across the Philippine archipelago, store proportionally less carbon in soils but remain ecologically significant (Rahman et al., 2024). These differences are reflected in the country-specific emission factors applied in this analysis.



Map 1. Net mangrove area change (%) by country relative to the 1996 baseline. Myanmar shows the highest proportional loss (dark red), while Brunei and Timor-Leste remain near stable or show marginal gains (blue tones). Source: GMW v3.0 (Bunting et al., 2022).

3. Data and Methods

3.1 Data source

Mangrove extent data were obtained from Global Mangrove Watch version 3.0 (GMW v3.0; Bunting et al., 2022), accessed through the Google Earth Engine (GEE) platform. GMW v3.0 maps global mangrove distribution at 100-metre spatial resolution using L-band Synthetic Aperture Radar (SAR) mosaics from the Japan Aerospace Exploration Agency (JAXA), covering eleven epochs: 1996, 2007, 2008, 2009, 2010, 2015, 2016, 2017, 2018, 2019, and 2020. The dataset is the most comprehensive time series of global mangrove extent currently available and is widely used in regional and global assessments (Rahman et al., 2024; Ximenes et al., 2023).

3.2 Change analysis

National mangrove extent was extracted by country using vector boundaries in GEE for each of the eleven epochs. Gross loss and gross gain were computed as the total area converted from mangrove to non-mangrove and from non-mangrove to mangrove, respectively, between the first (1996) and last (2020) epochs. Net change was calculated as gross gain minus gross loss. Percentage net change was expressed relative to the 1996 baseline extent. An annualised rate of change (ha yr^{-1}) was derived by dividing net change by the 24-year study period.

3.3 Trend analysis

The Mann-Kendall non-parametric test was applied to each country's eleven-epoch area time series to detect monotonic trends. This test was selected because GMW v3.0 epochs are not evenly spaced in time (a single

observation in 1996, annual coverage from 2007 to 2010, then annual coverage from 2015 to 2020), which violates assumptions of parametric trend tests. Mann-Kendall is robust to non-normal distributions and to irregular sampling intervals (Kendall, 1975). Trends were considered statistically significant at $p < 0.05$. Sen's Slope was calculated as the median of all pairwise slopes between epochs, providing a robust estimate of the median annual rate of change (ha yr^{-1}) that is resistant to outliers (Sen, 1968). Both tests were implemented in R using the trend package.

3.4 Blue carbon emissions

Blue carbon emissions were estimated by multiplying national net mangrove loss (ha) by country-specific emission factors ($\text{Mg CO}_2\text{e ha}^{-1}$) from Hamilton and Friess (2018). These factors integrate aboveground biomass, belowground biomass, and soil organic carbon to one metre depth following the IPCC Wetlands Supplement framework (Hiraishi et al., 2014). They represent the fraction of the whole-ecosystem carbon stock that is estimated to be mineralised and emitted upon deforestation, rather than treating the entire standing stock as immediately released. This approach differs from Rahman et al. (2024), who derived emission factors directly from the CO_2 -equivalent density values available on the Global Mangrove Alliance portal, effectively assuming complete stock-to-emission conversion. The present method is considered more conservative and more consistent with IPCC reporting standards. Country-specific carbon densities (Mg C ha^{-1}) and the corresponding CO_2e densities used in this analysis are summarised in Table 2.

Key Finding: Emission factors from Hamilton and Friess (2018) are calibrated against IPCC Wetlands Supplement standards and apply country-specific carbon densities, rather than a single regional average. This distinguishes the blue carbon estimates presented here from those in Rahman et al. (2024) and provides a basis for comparison.

4. Results

4.1 Regional mangrove extent and temporal trend

Total mangrove extent across the nine countries declined from 5.00 Mha in 1996 to 4.67 Mha in 2020, a net loss of approximately 330,000 ha representing 6.6 percent of the 1996 baseline. The trajectory shown in Figure 1 is not linear. The steepest decline occurred between 1996 and 2010, when the regional total dropped from 5.00 to 4.70 Mha, a loss of approximately 300,000 ha in fourteen years. From 2015 onward the rate of decline slowed markedly, with the regional extent stabilising around 4.64 to 4.67 Mha through to 2020. This deceleration is consistent with the expansion of national restoration programmes in Indonesia, Vietnam, and Myanmar following the 2015 Paris Agreement (Arifanti et al., 2022; Tinh et al., 2022).

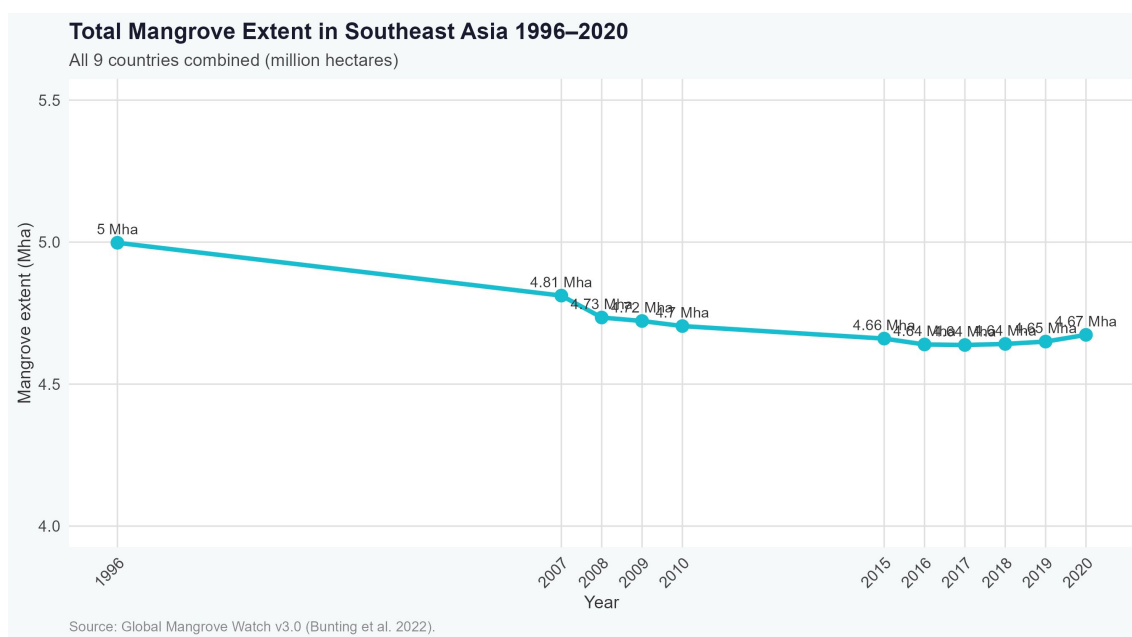


Figure 1. Total mangrove extent in Southeast Asia (nine countries combined), 1996 to 2020. The sharpest decline occurred between 1996 and 2010; the rate of loss slowed considerably after 2015. Source: GMW v3.0 (Bunting et al., 2022).

4.2 National net change and loss-gain dynamics

Table 1 summarises net change statistics for each of the nine countries. Indonesia recorded the largest absolute net loss at 210,463 ha (6.9 percent of its 1996 extent), reflecting decades of conversion to aquaculture, oil palm, and agricultural uses across Kalimantan and Sumatra. Myanmar shows the greatest proportional loss at 8.7 percent (49,850 ha), driven predominantly by rice agriculture expansion in Rakhine State (Richards and Friess, 2016). Vietnam lost 15,536 ha (8.0 percent), with loss concentrated before 2015 and a measurable recovery thereafter. Cambodia, Thailand, and Malaysia show moderate losses in the range of 1.9 to 4.0 percent. Brunei is the only country with a confirmed net gain, albeit marginal at 9.5 ha (0.07 percent), attributable to Integrated Coastal Zone Management policies implemented since 1993 (Islam et al., 2018). Timor-Leste shows a negligible net loss of 5.6 ha (0.5 percent).

Country	Baseline 1996 (ha)	Gross Loss (ha)	Gross Gain (ha)	Net Change (ha)	Net Change (%)
Indonesia	3,066,800	242,352	31,888	-210,463	-6.9%
Myanmar	575,979	64,670	14,821	-49,850	-8.7%
Malaysia	537,167	17,604	7,268	-10,336	-1.9%
Thailand	259,494	14,861	5,599	-9,262	-3.6%
Philippines	285,547	21,513	10,690	-10,823	-3.8%
Cambodia	65,106	3,285	694	-2,591	-4.0%
Vietnam	193,125	20,466	4,930	-15,536	-8.0%
Brunei	13,407	180	190	+9	+0.1%
Timor-Leste	1,054	58	52	-6	-0.5%

Table 1. Summary of mangrove area change by country, 1996 to 2020. Source: GMW v3.0 (Bunting et al., 2022).

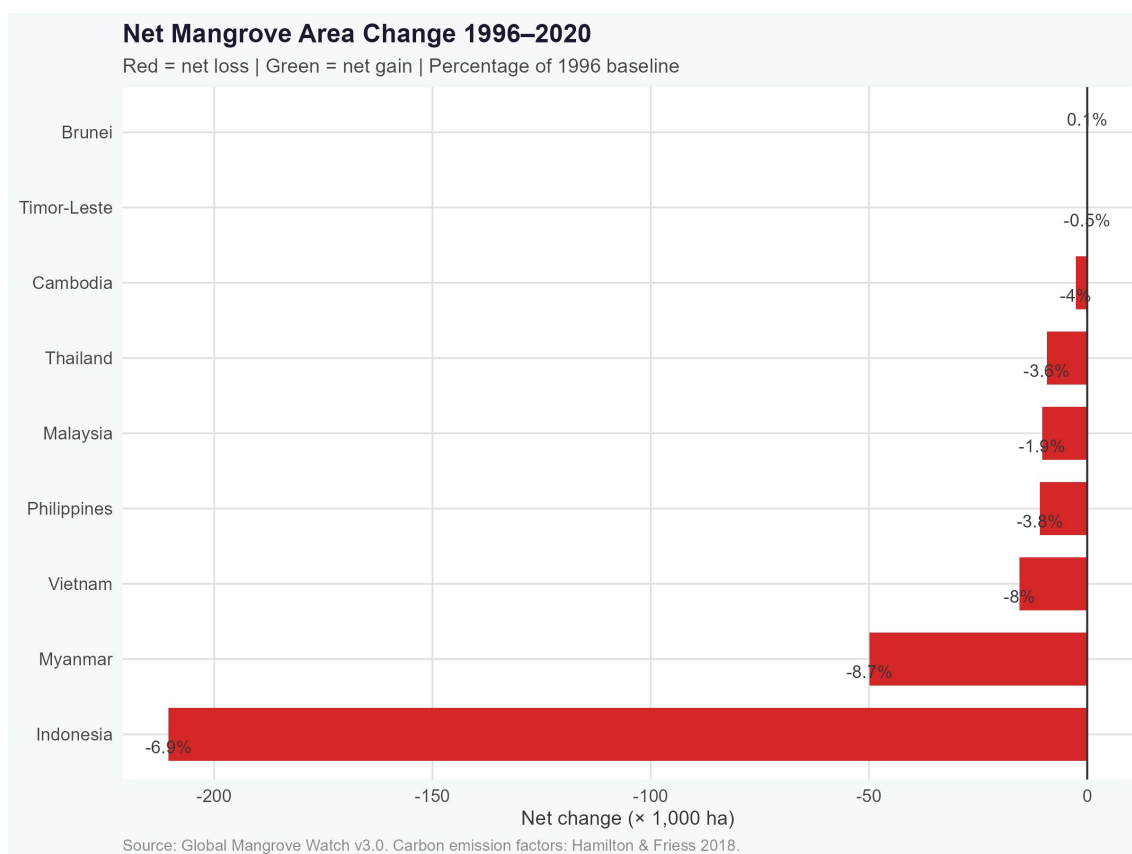


Figure 2. Net mangrove area change by country (ha × 1,000), 1996 to 2020. Red bars indicate net loss; the green bar (Brunei) indicates a marginal net gain. Percentages reflect change relative to the 1996 baseline. Source: GMW v3.0.

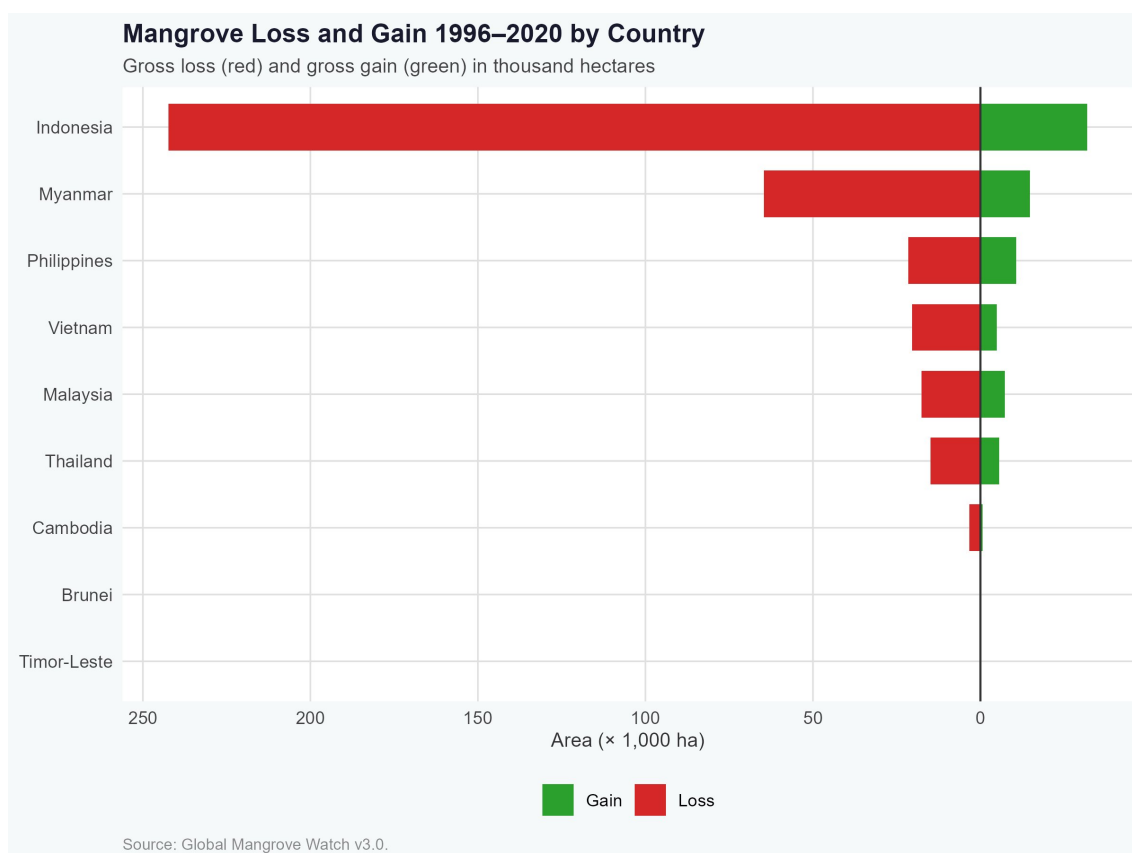


Figure 3. Gross loss (red) and gross gain (green) by country (ha × 1,000), 1996 to 2020. Indonesia and Myanmar show the largest absolute dynamics. Brunei and Timor-Leste show near-zero gross change. Source: GMW v3.0.

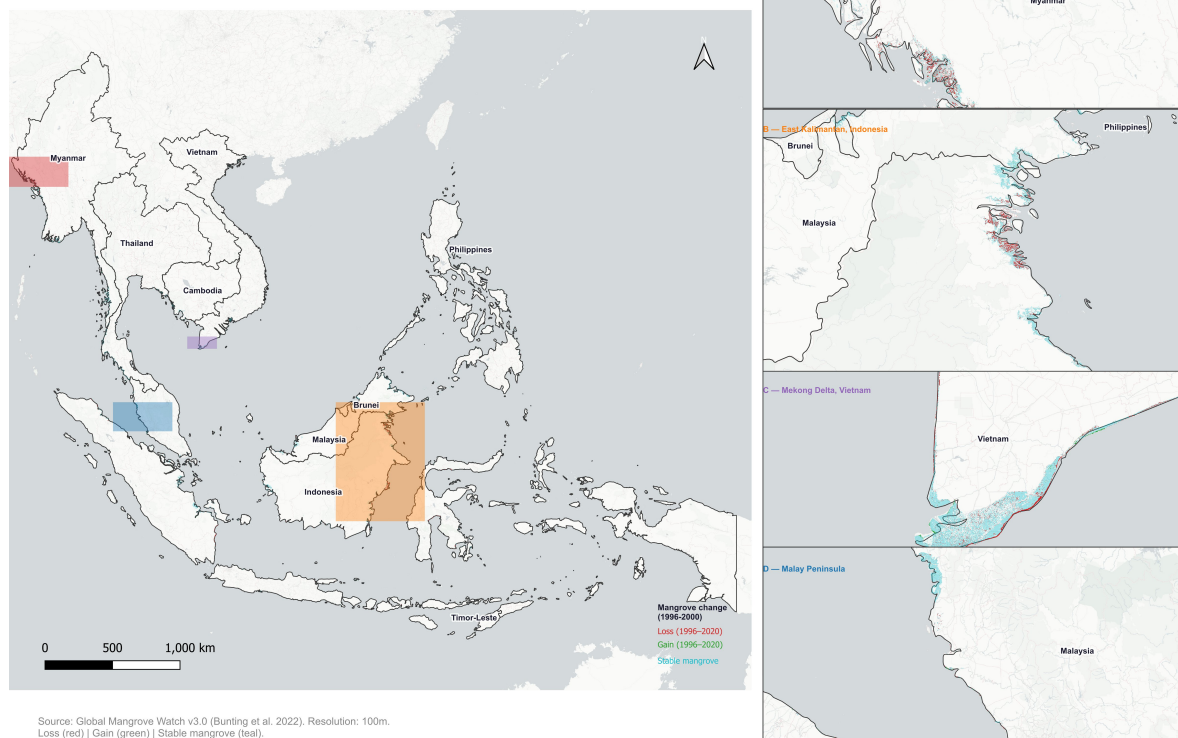
Figure 3 reveals that gross loss and gross gain occur simultaneously across most countries, reflecting the coexistence of ongoing deforestation and natural regeneration or active restoration. Indonesia shows the largest gross loss (242,352 ha) alongside a substantial gross gain (31,888 ha), suggesting that while conversion continues, partial recovery is occurring in abandoned areas or where protection efforts have taken hold. A similar pattern is visible in Myanmar and the Philippines, where gross gain represents approximately 23 and 50 percent of gross loss, respectively.

4.3 Spatial distribution of change

The spatial pattern of mangrove change shown in Map 2 highlights four sub-regional hotspots that concentrate a disproportionate share of regional loss. The Rakhine Coast of Myanmar (inset A) shows a narrow but intense band of red pixels along the coastline, consistent with the frontline of agricultural expansion documented by Richards and Friess (2016). East Kalimantan in Indonesia (inset B) shows a complex mosaic of loss and stable mangrove, reflecting the conversion to aquaculture ponds interspersed with areas under partial protection or recovery. The Mekong Delta in Vietnam (inset C) displays loss that is most intense in the southern tip of Ca Mau province, where shrimp farming pressure has been highest, alongside teal stable-mangrove areas that have benefited from restoration programmes (Tinh et al., 2022). The Malay Peninsula (inset D) shows a fragmented pattern with loss concentrated in northern Peninsular Malaysia where oil palm pressure is greatest.

Spatial Distribution of Mangrove Change in Southeast Asia (1996–2020)

Insets show key hotspots of mangrove loss and gain



Map 2. Spatial distribution of mangrove change across Southeast Asia, 1996 to 2020. Red pixels indicate loss, green pixels gain, and teal stable mangrove. Insets show key hotspots: (A) Rakhine Coast, Myanmar; (B) East Kalimantan, Indonesia; (C) Mekong Delta, Vietnam; (D) Malay Peninsula. Source: GMW v3.0 (Bunting et al., 2022).

4.4 Trend analysis: Mann-Kendall and Sen's Slope

Formal trend analysis identified statistically significant decreasing trends ($p < 0.05$) in four of the nine countries: Indonesia, Malaysia, Myanmar, and Vietnam (Table 2). Indonesia shows the strongest trend signal with a

Kendall's tau of -0.75 ($p = 0.002$) and a Sen's Slope of $-13,874 \text{ ha yr}^{-1}$, reflecting both the magnitude and consistency of loss across all eleven epochs. Malaysia exhibits a similarly strong tau of -0.71 ($p = 0.003$) and a slope of -908 ha yr^{-1} . Vietnam's trend (tau = -0.64, $p = 0.008$, slope = -793 ha yr^{-1}) captures the sustained pressure on the Mekong Delta and northern coastal zones through to approximately 2017, after which partial recovery becomes visible in the raw series. Myanmar's trend (tau = -0.53, $p = 0.029$, slope = $-3,326 \text{ ha yr}^{-1}$) is significant despite the strong post-2015 recovery visible in Figure 1, indicating that the net loss direction persists over the full period.

The remaining five countries do not reach statistical significance at $p < 0.05$. This does not necessarily indicate the absence of a directional trend; with only eleven observations the test has limited statistical power, and countries such as Cambodia and the Philippines show negative tau values (suggesting loss) that may reach significance with a longer or denser time series. Thailand and Timor-Leste show positive tau values, consistent with the slight recovery visible in their individual series after 2010, though these are also not statistically significant.

Country	Kendall's tau	p-value	Trend	Sen's Slope (ha/yr)
Indonesia	-0.745	0.002	Significant decrease	-13,874
Malaysia	-0.709	0.003	Significant decrease	-908
Vietnam	-0.636	0.008	Significant decrease	-793
Myanmar	-0.527	0.029	Significant decrease	-3,326
Cambodia	-0.345	0.161	Not significant	-155
Brunei	-0.309	0.213	Not significant	-5
Philippines	-0.236	0.350	Not significant	-309
Timor-Leste	+0.236	0.350	Not significant	+1
Thailand	+0.127	0.640	Not significant	+245

Table 2. Mann-Kendall trend test results and Sen's Slope estimates for mangrove area time series (1996 to 2020, $n = 11$ epochs). Asterisks (*) denote $p < 0.05$. Countries ordered by Kendall's tau.

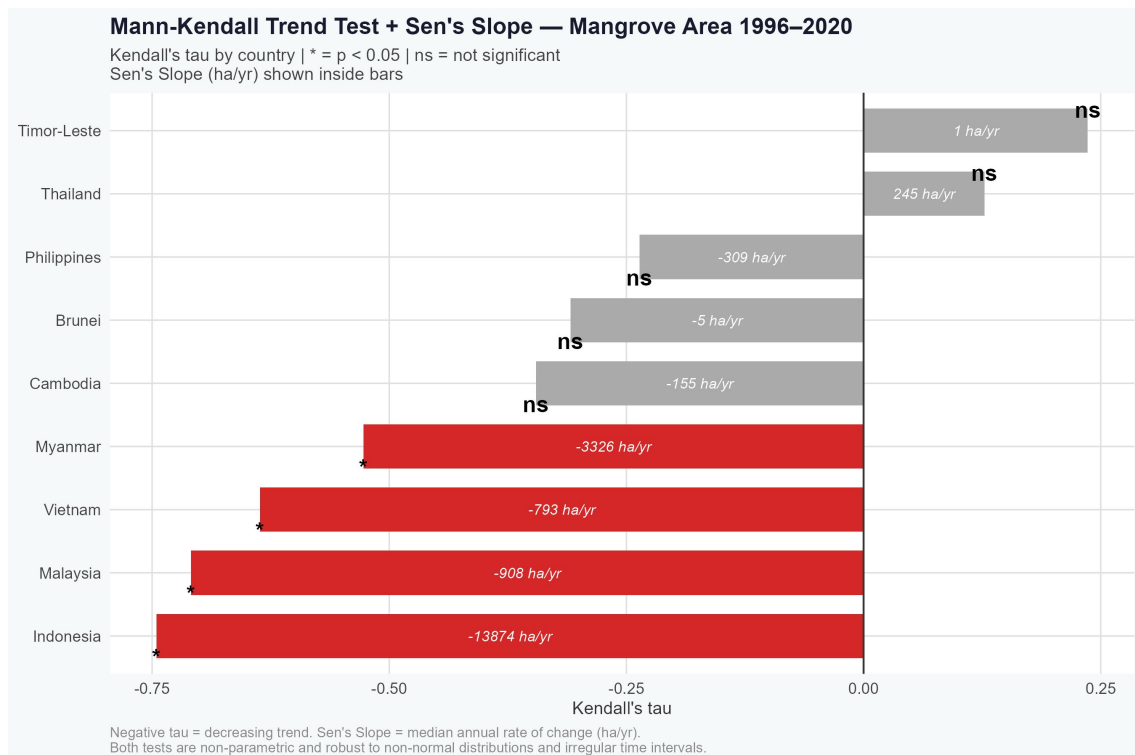


Figure 4. Mann-Kendall Kendall's tau and Sen's Slope (ha/yr, shown inside bars) for each country. Red bars indicate statistically significant decreasing trends ($p < 0.05$); grey bars are not significant. The slope magnitudes reflect the median annual rate of area change across all epoch pairs.

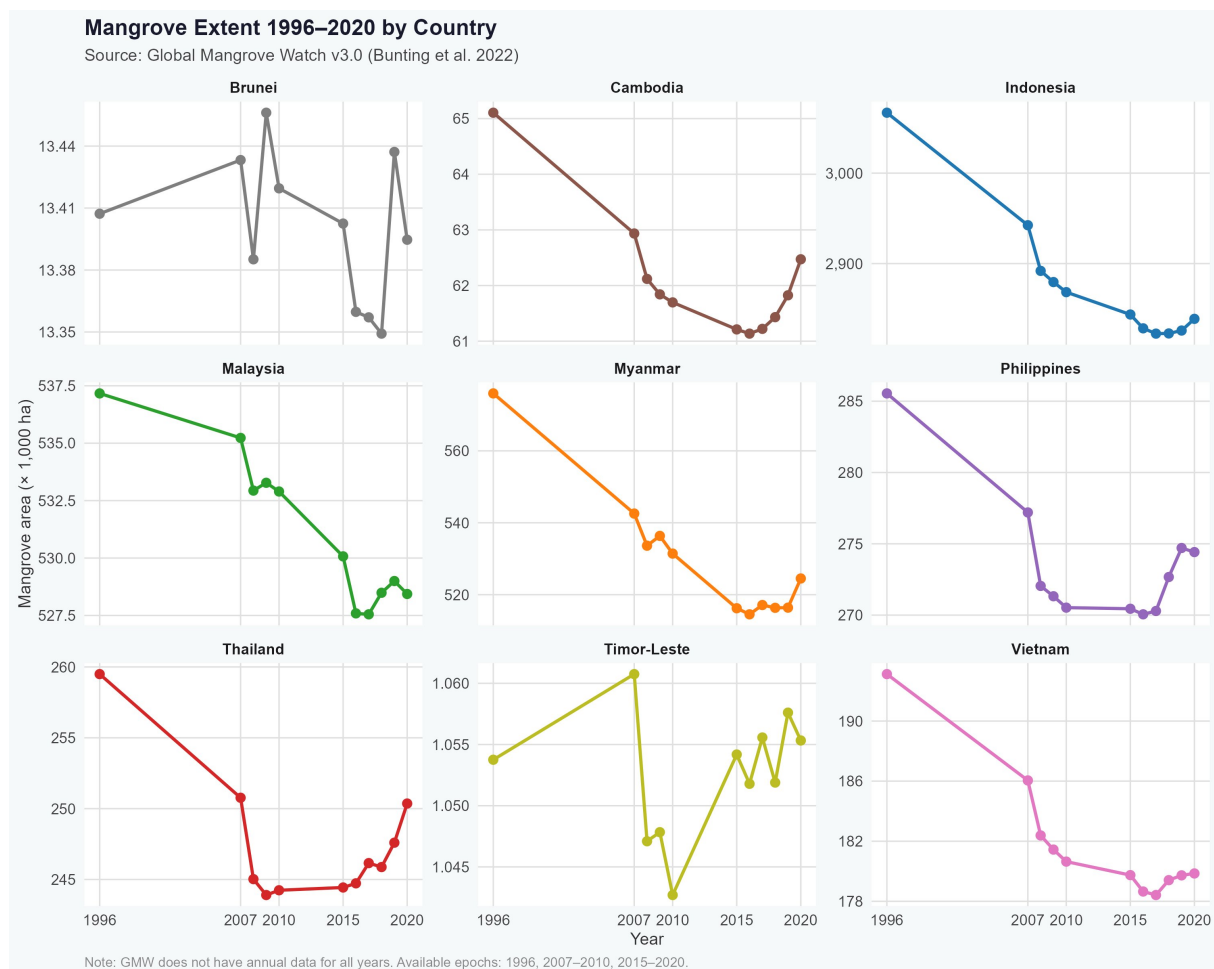


Figure 5. Mangrove extent time series by country (1996 to 2020). Each panel uses its own y-axis scale to highlight within-country dynamics. Note the steep post-2007 decline in Indonesia and Myanmar, and the partial recovery visible in several countries after 2015. Source: GMW v3.0 (Bunting et al., 2022).

4.5 Blue carbon emissions

Net blue carbon emissions associated with mangrove loss across the nine countries totalled approximately 582 Mt CO₂e between 1996 and 2020, equivalent to an average of about 24.2 Mt CO₂e yr⁻¹ over the study period. Indonesia dominates the regional total at 422 Mt CO₂e (72.5 percent), driven by the combination of the region's largest mangrove loss in absolute terms and among the highest country-specific carbon densities (546.7 Mg C ha⁻¹). Myanmar contributes 85 Mt CO₂e (14.6 percent), disproportionately large relative to its mangrove extent because loss was concentrated and sustained. Malaysia, the Philippines, Thailand, and Vietnam each contribute between 15 and 21 Mt CO₂e. Cambodia accounts for 3.2 Mt CO₂e. Brunei and Timor-Leste show negative emission values (net sequestration) consistent with their marginal net gains in mangrove area.

Country	C density (Mg C/ha)	CO ₂ e density (Mg/ha)	Net CO ₂ e (Mt)	Annual CO ₂ e (Mt/yr)
Indonesia	546.7	2,006.4	422.3	17.6
Myanmar	464.9	1,706.0	85.0	3.5
Malaysia	547.8	2,010.4	20.8	0.9
Philippines	506.1	1,857.4	20.1	0.8
Thailand	486.6	1,786.0	16.5	0.7
Vietnam	262.8	964.6	15.0	0.6
Cambodia	331.3	1,216.1	3.2	0.1
Timor-Leste	367.6	1,348.9	0.0	0.0
Brunei	480.0	1,761.5	-0.02	-0.001

Table 3. Blue carbon emission estimates by country (1996 to 2020). Carbon and CO₂e densities from Hamilton and Friess (2018). Negative values for Brunei and Timor-Leste indicate net carbon sequestration due to marginal mangrove gain.

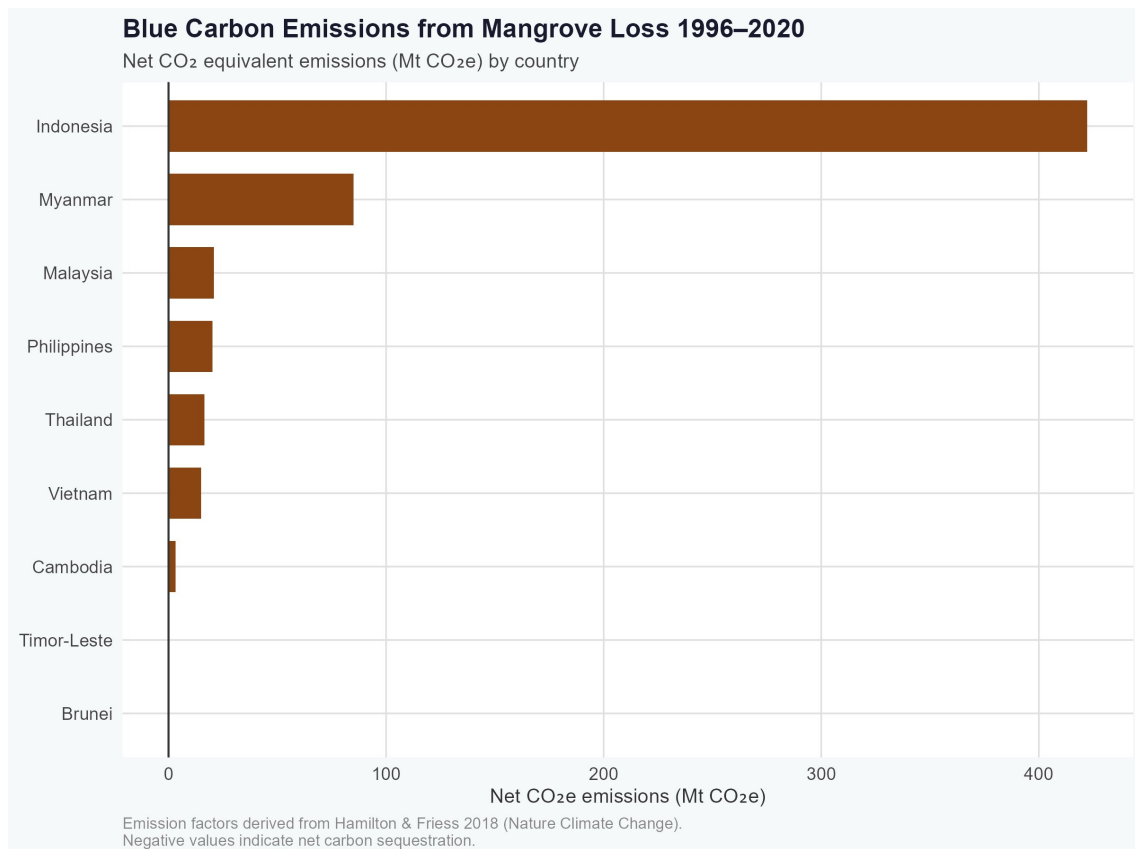


Figure 6. Net CO₂e emissions from mangrove loss by country (Mt CO₂e, 1996 to 2020). Negative values for Brunei and Timor-Leste reflect net carbon sequestration. Emission factors from Hamilton and Friess (2018).



Map 3. Spatial distribution of net blue carbon emissions by country (Mt CO₂e, 1996 to 2020). Circle size is proportional to total emissions. Indonesia (422 Mt CO₂e) and Myanmar (85 Mt CO₂e) account for approximately 87 percent of regional emissions. Emission factors from Hamilton and Friess (2018).

Key Finding: Indonesia alone is responsible for 72.5 percent of regional blue carbon emissions (422 Mt CO₂e). Together with Myanmar, the two countries account for approximately 87 percent of the regional total, making them the priority targets for any blue carbon-based conservation or payments-for-ecosystem-services mechanism in Southeast Asia.

5. Discussion

5.1 Scale and trajectory of loss in regional context

The net loss of approximately 330,000 ha documented here is larger than the figure reported by Rahman et al. (2024), who estimated a decline of roughly 250,000 ha (5.07 to 4.82 Mha). The difference reflects the exclusion of Singapore in the present analysis and likely differences in GEE extraction boundaries rather than contradictory data. Both studies draw from the same GMW v3.0 product and cover the same period; the estimates are therefore complementary rather than competing. The loss documented here is also substantially larger than the approximately 97,900 ha of net habitat loss estimated by Richards and Friess (2016) for the 2000 to 2012 period alone, though that study covered a shorter timeframe and used a different baseline dataset. Taken together, these figures confirm that mangrove loss in Southeast Asia over the past three decades has been persistent and large in absolute terms, even as the rate has slowed after 2015.

The deceleration visible in the regional trajectory after 2015 (Figure 1) is consistent with several concurrent policy developments. The Paris Agreement of 2015 prompted several Southeast Asian governments to incorporate mangrove conservation into their Nationally Determined Contributions (NDCs) (Murdiyarso et al., 2015; Arifanti et al., 2022). Indonesia's Peatland and Mangrove Restoration Agency committed to rehabilitating 600,000 ha of mangrove by 2024 (Rahman et al., 2024). Vietnam undertook restoration at a rate of approximately 2,237 ha yr⁻¹ between 2015 and 2020 (Tinh et al., 2022). Myanmar received support from the Global Green Growth Institute for targeted restoration in 2019 (Rahman et al., 2024). Whether this stabilisation represents a genuine turning point or a temporary plateau remains to be confirmed as post-2020 GMW data become available.

5.2 Country-level drivers

The spatial and national patterns documented here align closely with the driver analysis of Richards and Friess (2016), whose remote sensing classification showed that aquaculture accounts for 30 percent of regional mangrove deforestation, rice expansion for 22 percent (dominated by Myanmar), and oil palm for 16 percent (concentrated in Malaysia and Indonesia). Indonesia's dominant position in both absolute loss and blue carbon emissions reflects the scale of aquaculture-driven conversion in Kalimantan and Sulawesi, which continued to intensify through the early 2010s partly in response to government policies promoting Indonesia as the world's leading aquaculture producer (Richards and Friess, 2016; Arifanti et al., 2019).

Myanmar's proportional loss of 8.7 percent is the highest in the dataset and is concentrated in Rakhine State, where rice agriculture has expanded into formerly forested intertidal zones (Richards and Friess, 2016). The significant Mann-Kendall trend ($\tau = -0.53$, $p = 0.029$) confirms that this loss has been directional rather than episodic, even though the raw time series shows partial recovery after 2015 that may reflect abandonment of marginal agricultural plots rather than active restoration. This interpretation is consistent with Richards and Friess (2016), who noted that the Myanmar government provided few environmental safeguards for mangrove forests during the study period and that rice expansion was driven largely by village-level smallholder activity.

Vietnam's trajectory is particularly instructive. The significant decreasing trend ($\tau = -0.64$, $p = 0.008$) captures the sustained loss in the Mekong Delta and Ca Mau Province driven by shrimp farming (Hauser et al., 2017; Luo et al., 2022), while the raw time series (Figure 5) shows a measurable stabilisation after 2017. This is consistent with Tinh et al. (2022), who documented restoration of $2,237 \text{ ha yr}^{-1}$ in the Vietnamese Mekong Delta between 2015 and 2020. That recovery is insufficient to offset the historical loss captured by the trend test, but it represents a meaningful policy response that countries with greater loss trajectories have yet to match.

Malaysia's statistically significant trend ($\tau = -0.71$, $p = 0.003$) despite its comparatively moderate net loss percentage (1.9 percent) reflects the consistency of loss across all eleven epochs rather than any single dramatic event. Oil palm plantation expansion accounts for 38.2 percent of deforested mangrove area in Malaysia (Richards and Friess, 2016), a pressure that intensified through the 2000s and has been difficult to regulate given the fragmented governance of intertidal zones between marine and terrestrial agencies (Wever et al., 2012). Brunei's marginal net gain (9.5 ha over 24 years) stands in contrast to its neighbours and has been attributed to the Integrated Coastal Zone Management programme implemented since 1993 (Islam et al., 2018; Rahman et al., 2024), demonstrating that policy coherence at the national scale can maintain mangrove extent even under regional deforestation pressure.

5.3 Methodological contribution: trend formalisation

A key distinction between this report and prior regional assessments is the application of a formal statistical framework to characterise national-level trends. Richards and Friess (2016) described trends verbally based on mapped change rates. Rahman et al. (2024) reported deforestation rates and temporal breakdowns but did not test whether these trajectories represent statistically significant monotonic trends. The Mann-Kendall test adds a layer of rigour that is particularly relevant for policy applications: a country with a statistically significant decreasing trend over a 24-year period presents stronger evidence of systemic loss pressure than one where change may be explained by natural variability or measurement uncertainty in the remote sensing product.

The finding that only four of nine countries show significant trends also underscores an important limitation of the GMW epoch structure. With eleven observations spread unevenly across 24 years, the Mann-Kendall test has limited statistical power, particularly for countries with small mangrove extents or high inter-epoch variability such as Timor-Leste and Brunei. Countries that appear stable in the statistical sense may nonetheless be experiencing ecologically meaningful change at sub-national scales that the national aggregate masks. Future assessments with annual GMW data from 2015 onwards will substantially improve the power of this type of analysis.

5.4 Blue carbon emissions: methodological comparison with Rahman et al. (2024)

The total regional emission estimate of 582 Mt CO_2e presented here differs from the emission figures in Rahman et al. (2024), whose approach derives emission factors directly from the CO_2 -equivalent density values available on the Global Mangrove Alliance portal, effectively treating the entire standing stock as emitted upon deforestation. This assumption is likely to overestimate emissions for land-use conversions where a portion of soil carbon remains sequestered, for example where mangroves are converted to aquaculture ponds that retain waterlogged, reducing conditions in deeper soil horizons (Pendleton et al., 2012; Lovelock et al., 2017).

The Hamilton and Friess (2018) emission factors used here follow the IPCC Wetlands Supplement framework (Hiraishi et al., 2014), which recommends a one-metre soil depth assumption for emissions accounting under the rationale that it is the upper metre of soil that is most vulnerable to disturbance and remineralisation. Country-specific factors range from $964.6 \text{ Mg CO}_2\text{e ha}^{-1}$ for Vietnam, whose open-coast mangroves have relatively lower soil carbon accumulation, to $2,010.4 \text{ Mg CO}_2\text{e ha}^{-1}$ for Malaysia and $2,006.4 \text{ Mg CO}_2\text{e ha}^{-1}$ for

Indonesia, whose estuarine and bay-type ecosystems accumulate far greater soil carbon stocks (Rahman et al., 2024; Murdiyarso et al., 2015). This spatial heterogeneity in carbon density has real policy implications: a hectare of mangrove lost in Indonesian Kalimantan carries roughly twice the climate penalty of a hectare lost in Vietnam's open-coast zones.

The Indonesia-dominant pattern (72.5 percent of regional emissions from 63 percent of regional 1996 extent) is consistent with Hamilton and Friess (2018), who found that Indonesia alone held over 30 percent of global mangrove carbon stocks in 2012, and with Murdiyarso et al. (2015), who estimated that halting Indonesian mangrove deforestation could reduce the country's land-use sector emissions by 10 to 31 percent. Myanmar's disproportionate contribution (85 Mt CO₂e from 8.7 percent proportional loss) reflects the combination of sustained loss and moderate carbon density (464.9 Mg C ha⁻¹), which is lower than Indonesia's but still substantial.

5.5 Policy implications

The four countries with statistically significant decreasing trends, Indonesia, Malaysia, Myanmar, and Vietnam, are also among the five largest emitters of blue carbon in the dataset. This alignment is not coincidental: persistent, directional mangrove loss generates cumulative emissions that accumulate over decades. Under the NDC frameworks submitted to the UNFCCC, all four countries have included land-use change in their mitigation targets, and Indonesia, Vietnam, and Malaysia have explicitly identified mangrove conservation and restoration as instruments for meeting those targets (Arifanti et al., 2022). The blue carbon figures presented here can serve as a baseline against which to assess progress, provided the methodological caveats regarding emission factor selection and the limitations of remote sensing-based change detection are taken into account.

Payments for ecosystem services mechanisms, including REDD+ and voluntary carbon markets, have increasingly looked to blue carbon as a source of verifiable, tradeable credits (Mcleod et al., 2011; Hamilton and Friess, 2018). The sub-national carbon stock data provided by Hamilton and Friess (2018) enable a more granular assessment of where conservation investment would generate the greatest climate return per hectare, pointing to the estuarine and bay-type mangrove systems of Indonesian Kalimantan and Sumatra as the highest-priority sites in the region.

6. Conclusions

Southeast Asia lost approximately 330,000 ha of mangrove between 1996 and 2020, a net decline of 6.6 percent from the 1996 baseline of 5.0 Mha. The loss was most intense during the 1996 to 2010 period and decelerated markedly after 2015, a pattern consistent with the expansion of national restoration programmes and the integration of mangrove conservation into post-Paris Agreement climate commitments across the region.

Formal trend analysis using the Mann-Kendall test identified statistically significant decreasing trends in four countries: Indonesia (tau = -0.75), Malaysia (tau = -0.71), Vietnam (tau = -0.64), and Myanmar (tau = -0.53). The remaining five countries do not show significant trends at $p < 0.05$, though this may partly reflect the limited statistical power of the eleven-epoch GMW time series rather than the absence of directional change.

Blue carbon emissions associated with mangrove loss totalled approximately 582 Mt CO₂e, with Indonesia accounting for 72.5 percent (422 Mt CO₂e) and Myanmar for a further 14.6 percent (85 Mt CO₂e). These figures are derived using country-specific emission factors from Hamilton and Friess (2018) and are not directly comparable to estimates from studies using different emission factor methodologies, including Rahman et al. (2024). Brunei and Timor-Leste show marginal net sequestration, confirming that effective coastal zone

management can maintain or modestly expand mangrove extent even under regional deforestation pressure.

This report is offered as a technical complement to the body of research on Southeast Asian mangrove dynamics, particularly the foundational work of Richards and Friess (2016) on deforestation drivers, Hamilton and Friess (2018) on carbon stocks, and Rahman et al. (2024) on regional blue carbon dynamics. It does not seek to refute or supersede those contributions, which carry greater analytical depth, field validation resources, and methodological rigour. The value of the present analysis lies in extending the temporal coverage to the full 1996 to 2020 period, adding a formal statistical characterisation of national trends, and applying country-specific emission factors to produce an independently derived set of baseline emission estimates.

Limitations

Several limitations should be considered when interpreting these results. Remote sensing data from GMW v3.0, although based on SAR imagery that is less susceptible to cloud cover than optical sensors, is not immune to artefacts. Persistent cloud cover during image acquisition windows, particularly in the humid tropics, can introduce errors in the detection of change at the margins of mangrove patches. The 100-metre spatial resolution of the product means that small-scale gains or losses, and degradation that does not result in complete canopy removal, are not captured.

The eleven-epoch structure of GMW v3.0 imposes constraints on trend analysis. The large gap between 1996 and 2007 means that dynamics during that eleven-year window are represented by only two data points, potentially missing important sub-decadal variability. The limited number of observations ($n = 11$) reduces the statistical power of the Mann-Kendall test, particularly for countries with small extents or high inter-epoch variability.

The blue carbon emission estimates assume that net mangrove loss is equivalent to outright deforestation and apply national average emission factors from Hamilton and Friess (2018). In reality, emissions depend on the specific land use that replaces mangrove, the depth of soil disturbance, and the hydrological conditions of the converted site (Pendleton et al., 2012; Lovelock et al., 2017). Sub-national variation in carbon density, which can be substantial within a country like Indonesia, is not captured at the resolution of this analysis. No field validation of the change detection results was conducted; the estimates should therefore be treated as remote sensing-based approximations that require ground-truthing before use in formal carbon accounting or REDD+ baseline setting.

Key Finding: The findings presented here are consistent with the broader literature on Southeast Asian mangrove dynamics and are intended to support, not replace, the more detailed assessments produced by specialists in the field. Future work combining annual GMW data with sub-national carbon stock maps and field verification would substantially improve the precision of both trend analysis and emission estimates.

References

- Arifanti, V.B., Kauffman, J.B., Subarno, Ilman, M., Tosiani, A., Novita, N. (2022). Contribution of mangrove conservation and restoration to climate change mitigation in Indonesia. *Global Change Biology*, 28, 4523–4538.
- Arifanti, V.B., Kauffman, J.B., Hadriyanto, D., Murdiyarso, D., Diana, R. (2019). Carbon dynamics and land use carbon footprints in mangrove-converted aquaculture: the case of the Mahakam Delta, Indonesia. *Forest Ecology and Management*, 432, 17–29.

- Bunting, P., Rosenqvist, A., Hilarides, L., Lucas, R.M., Thomas, N., Tadono, T., Worthington, T.A., Spalding, M., Murray, N.J., Rebelo, L.M. (2022). Global mangrove extent change 1996–2020: Global Mangrove Watch Version 3.0. *Remote Sensing*, 14, 3657.
- Donato, D.C., Kauffman, J.B., Murdiyarso, D., Kurnianto, S., Stidham, M., Kanninen, M. (2011). Mangroves among the most carbon-rich forests in the tropics. *Nature Geoscience*, 4, 293–297.
- Giri, C., Ochieng, E., Tieszen, L.L., Zhu, Z., Singh, A., Loveland, T., Masek, J., Duke, N. (2011). Status and distribution of mangrove forests of the world using earth observation satellite data. *Global Ecology and Biogeography*, 20, 154–159.
- Hamilton, S., Friess, D.A. (2018). Global carbon stocks and potential emissions due to mangrove deforestation from 2000 to 2012. *Nature Climate Change*, 8, 240–244.
- Hauser, L.T., Vu, G.N., Nguyen, B.A., et al. (2017). Uncovering the spatio-temporal dynamics of land cover change and fragmentation of mangroves in the Ca Mau peninsula, Vietnam using multi-temporal SPOT satellite imagery (2004–2013). *Applied Geography*, 86, 197–207.
- Hiraishi, T., Krug, T., Tanabe, K., Srivastava, N., Baasansuren, J., Fukuda, M., Troxler, T.G. (2014). *2013 Supplement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories: Wetlands*. IPCC, Switzerland.
- Hochard, J.P., Hamilton, S., Barbier, E.B. (2019). Mangroves shelter coastal economic activity from cyclones. *Proceedings of the National Academy of Sciences*, 116, 12232–12237.
- Islam, S.N., Rahman, N.H.H.A., Reinstadtler, S., Aladin, M.N.A.B. (2018). Assessment and management strategies of mangrove forests alongside the Mangsalut River Basin (Brunei Darussalam). In Makowski, C., Finkl, C. (Eds.), *Threats to Mangrove Forests*. Springer.
- Kendall, M.G. (1975). *Rank Correlation Methods*. 4th edition. Charles Griffin, London.
- Lovelock, C.E., Fourqurean, J.W., Morris, J.T. (2017). Modeled CO₂ emissions from coastal wetland transitions to other land uses: tidal marshes, mangrove forests, and seagrass beds. *Frontiers in Marine Science*, 4, 143.
- Luo, J., Sun, Z., Lu, L., Xiong, Z., Cui, L., Mao, Z. (2022). Rapid expansion of coastal aquaculture ponds in southeast Asia: patterns, drivers, and impacts. *Journal of Environmental Management*, 315, 115100.
- Mcleod, E., Chmura, G.L., Bouillon, S., Salm, R., Bjork, M., Duarte, C.M., et al. (2011). A blueprint for blue carbon: toward an improved understanding of the role of vegetated coastal habitats in sequestering CO₂. *Frontiers in Ecology and the Environment*, 9, 552–560.
- Murdiyarso, D., Purbopuspito, J., Kauffman, J.B., Warren, M.W., Sasmito, S.D., Donato, D.C., Kurnianto, S., Stidham, M., Kanninen, M. (2015). The potential of Indonesian mangrove forests for global climate change mitigation. *Nature Climate Change*, 5, 1089–1092.
- Pendleton, L., Donato, D.C., Murray, B.C., Crooks, S., Jenkins, W.A., Sifleet, S., et al. (2012). Estimating global blue carbon emissions from conversion and degradation of vegetated coastal ecosystems. *PLoS ONE*, 7(9), e43542.
- Primavera, J.H., Friess, D.A., Van Laveren, H., Lee, S.Y. (2019). The mangrove ecosystem. In *World Seas: An Environmental Evaluation*, pp. 1–34. Elsevier.
- Rahman, et al. (2024). Mangrove ecosystems in Southeast Asia region: Mangrove extent, blue carbon potential and CO₂ emissions in 1996–2020. *Science of the Total Environment*, 915, 170052.
- Richards, D.R., Friess, D.A. (2016). Rates and drivers of mangrove deforestation in Southeast Asia, 2000–2012. *Proceedings of the National Academy of Sciences*, 113(2), 344–349.
- Sen, P.K. (1968). Estimates of the regression coefficient based on Kendall's tau. *Journal of the American Statistical Association*, 63, 1379–1389.
- Sharma, S., Ray, R., Martius, C., Murdiyarso, D. (2023). Carbon stocks and fluxes in Asia-Pacific mangroves: current knowledge and gaps. *Environmental Research Letters*, 18, 044002.
- Spalding, M., Kainuma, M., Collins, L. (2010). *World Atlas of Mangroves*. Earthscan, London.
- Tinh, P.H., MacKenzie, R.A., Hung, T.D., et al. (2022). Mangrove restoration in Vietnamese Mekong Delta during 2015–2020: achievements and challenges. *Frontiers in Marine Science*, 9, 1043943.

- Wever, L., Glaser, M., Gorris, P., Ferrol-Schulte, D. (2012). Decentralization and participation in integrated coastal management: policy lessons from Brazil and Indonesia. *Ocean and Coastal Management*, 66, 63–72.
- Ximenes, A.C., Cavanaugh, K.C., Arvor, D., et al. (2023). A comparison of global mangrove maps: assessing spatial and bioclimatic discrepancies at poleward range limits. *Science of the Total Environment*, 860, 160380.